MIL-C-63296 (AR)
22 January 1980
SUPERSEDING
FA-PD-MI-2523, Rev. 1
1 July 1964

MILITARY SPECIFICATION

CASE, CARTRIDGE, 105MM, M115B1

This specification is approved for use by US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.-This specification contains requirements not covered by the drawings and provides quality assurance provisions for the fabrication of a case, cartridge, 105mm, M115B1.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents.-The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

FSC: 1315

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS

ARMY

7548235 - Cartons, Ammunition, for Interplant Shipment of Steel Cartridge Cases, Empty 37MM to 120MM, Inclusive
8663760 - Gage, Chamber
10520214 - Case, Cartridge, 105MM, M115B1

(Copies of specifications, standards, drawings and publications listed above or in the item detail specification required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B46.1 - Surface Texture,

Application for copies should be addressed to:

American National Standards Institute (ANSI), 1430 Broadway, New York, NY 10018.

ASTM Methods -AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM B117 - Standard Method of Salt Spray (Fog) Testing
ASTM E8 - Standard Methods of Tension Testing of Metallic Materials
ASTM E18 - Standard Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials
ASTM E381 - Standard Method for Rating Macroetched Steel

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

3. REQUIREMENTS

3.1 General.-The cartridge case shall comply with the requirements of the applicable drawings, reference specifications and the following:

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3.2 Material.-The supplier shall be responsible for selecting from the classification list contained in the raw material specifications, the surface and physical condition of the raw material intended for use in manufacture.

3.3 Mechanical properties.-The mechanical properties specified on the applicable drawings and referenced specifications shall be uniform throughout each section.

3.3.1 Thermal treatment.-The entire cartridge case shall be subjected to a thermal treatment after the heading operation by heating above the upper critical temperature point, quenching and tempering to the desired properties.

3.3.2 Stress relief.-The cartridge case shall be adequately stress relieved at the times and temperatures suitable for development of desired properties.

3.4 Protective coating.-Protective coating as specified on the applicable drawings and reference specifications shall apply to all surfaces of the cartridge case regardless of contour configuration and shall be uniform and continuous. Scratches exposing base metal will not be permitted.

3.5 Maximum profile and alignment.-Maximum profile and alignment of the cartridge case shall comply with the requirements of the applicable drawing.

3.6 Functional.-The cartridge case shall be subjected to a functional test in the following phases:

a. Excess pressure phase
b. Cold temperature phase
c. Excess, excess pressure phase (for information only).

3.6.1 Obturation.-The cartridge case shall effectively seal the breech of the weapon against leakage of gases generated during firing.

3.6.2 Soundness.-After firing, the cartridge case shall be free of cracks or splits.

3.6.3 Extraction-After firing, the cartridge case shall be extracted by the service extractors when the weapon is operated in the normal manner. (See 6.4).

3.6.4 Primer retention.-The cartridge case shall retain its primer during firing and extraction.

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3.7 First article inspection.-This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.8 Workmanship.-The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.8.1 Metal defects.-The cartridge case shall be free from cracks, blisters, seams, laminations, deep scratches or porosity.

3.8.2 Flow of metal.-The flow of metal in the section of the cartridge case joining the head and sidewall shall follow the direction of the contour of the interior curved surface without presence of folds, cold shuts or roughness.

3.8.3 Threads.-Threads shall be full and undamaged for the entire minimum length or depth as required by the applicable drawing and specifications.

3.8.4 Burr.-No cartridge case shall have a burr.

3.8.5 Foreign matter.-No cartridge case shall contain chips, dirt, grease, oil, rust, corrosion, or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections.-The following types of inspection shall be conducted on this item:

a. First Article inspection
b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission.-The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article shall consist of the following items in sample quantities as indicated.

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Part Description	Drawings	Quantity
Case, Cartridge (Uncoated)	10520214	20
Case, Cartridge (Coated)	10520214	254*

*All samples must meet the requirements for thermal treatment and high stress expansion tests as performed in the uncoated state.

4.3.2 Inspection to be performed.-See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection.-See MIL-A-48078.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63296 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Case Cartridge 105MM, M115B1				10520214
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Cartridge Case - Prior to Coating				
	Examination for Defects	20		3.1	4.4.2.1
	Hardness	20		3.3	4.5.3
	Macro Examination	4		3.8.2	4.5.5
	Tension Test (Info. Only)	4		3.3	4.5.4
	Cartridge Case - After Coating				
	Examination for Defects	254		3.1	4.4.2.2
	Thickness of Coating	2		3.4	4.5.6.1
	Salt Spray	2		3.1	4.5.6.2
	Ballistic Test				
	Excess Pressure Phase	100	Note 1	3.6	4.5.9.1.1
	Cold Temperature Phase	50	Note 1	3.6	4.5.9.1.2
	Excess, Excess Pressure Phase (Info. Only)	100		3.6	4.5.9.1.3

NOTES:
1. Acceptance Rejection Criteria: See Table II.

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4.4 Quality conformance inspection

4.4.1 Lot formation.-Lots shall comply with the lot formation provisions of MIL-A-48078. In addition one lot of cartridge cases shall consist of not more than 25,000 units.

4.4.2 Examination.-See MIL-A-48078.

a. Sampling plans.-Unless otherwise specified in the Classification of Defects and test tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

b. Tool control dimensions.-Dimensions marked tool control (see Dwg. 10520214) shall be gaged during First Article testing and prior to each 8 hours shift production on each machine in operation and prior to any tooling change. Where destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted provided the contractor has established correlation between the tool dimensions and the component dimensions prior to the start of production. The contractor shall be responsible for maintaining adequate surveillance to assure that drawing requirements are being met throughout the life of the tool. Corrective action will immediately be instituted by the contractor whenever a defective tool control dimension is found. All material produced prior to periodic inspection will be rejected and screened whenever a defective tool control dimension is found.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63296 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.1	Case, Cartridge - Prior to Coating				10520214
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101.	Thermal treatment.		100% (c)	3.3.1	4.5.1
102.	High stress expansion test.		100% (c&d)	3.1	4.5.2
103.	Cracks.		100% (c)	3.8.1	4.5.7.1
104.	Metal defective.		100% (c)	3.8.1	4.5.7
105.	Hardness.	4	(a)	3.3	4.5.3
106.	Tension test (info only).	4	(b)	3.3	4.5.4
107.	Macro Examination	4	(a)	3.8.2	4.5.5
108.	Diameter of flange, min.		0.40%	3.1	Gage
109.	Diameter of mouth		0.40%	3.1	Gage
110.	Length of mouth diameter, min.		0.40%	3.1	Gage
111.	Inside diameter before taper at mouth end.		0.40%	3.1	Gage
112.	True position of inside diameter before taper at mouth end with mouth diameter.		0.40%	3.1	Gage
113.	Thickness of flange, min.		0.40%	3.1	Gage
114.	Thickness of base, min.		0.40%	3.1	Gage
115.	Wall thickness at first datum from base.		0.40%	3.1	Gage
116.	Wall thickness at second datum from base.		0.40%	3.1	Gage
117.	Wall thickness at third datum from base.		0.40%	3.1	Gage
118.	Wall thickness at mouth		0.40%	3.1	Gage
119.	Contour of inside of base and sidewall not smooth		0.40%	3.1	Gage

NOTES:
a. Reject on 1 Defect.
b. Samples for the hardness test may be used in this test.
c. All non-compliance units must be removed from the lot.
d. Evidence of cracks in any portion of the case prior to or after tapering shall be cause for rejection of case.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63296 (AR)

PARAGRAPH	TITLE	SHEET 2 OF 2	DRAWING NUMBER
4.4.2.1	CONTINUATION SHEET		10520214
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201.	Diameter of body under flange, min.		0.65%	3.1	Gage
202.	Location of first basic diameter from flange, min.		0.65%	3.1	Gage
203.	Location of second basic diameter from flange, min.		0.65%	3.1	Gage
204.	Location of third basic diameter from flange, min.		0.65%	3.1	Gage
205.	Location of fourth basic diameter from flange, min.		0.65%	3.1	Gage
206.	Radii or chamfers, missing or incorrect		0.65%	3.1	Visual
207.	Surface finish, improper		0.65%	3.1	Visual
208.	Evidence of poor workmanship		0.65%	3.8	Visual
209.	Length total, min.		0.65%	3.1	Gage

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.2	Case, Cartridge - After Coating		10520214
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Cracks		100%(a)	3.8.1	4.5.7.1
102	Metal defective		100%(a)	3.8.1	4.5.7
103	Thickness of coating	(See 4.5.6)		3.4	4.5.6.1
104	Salt spray	(See 4.5.6)		3.1	4.5.6.2
105	Maximum profile and alignment		100%	3.5	4.5.8
106	Excess pressure ballistic test	(See 4.4.3.1.2)		3.6	4.5.9.2.1
107	True position of pitch diameter of thread with body		0.40%	3.1	Gage
108	True position of primer hole counterbore with pitch diameter of thread		0.40%	3.1	Gage
109	True position of primer hole thread relief with pitch diameter of thread		0.40%	3.1	Gage
110	Pitch diameter of thread, max.		0.40%	3.1	Gage
111	Minor diameter of thread, max.		0.40%	3.1	Gage
112	Length total, max.		0.40%	3.1	Gage
113	Diameter of primer hole counterbore		0.40%	3.1	Gage
114	Depth of primer hole counterbore, min.		0.40%	3.1	Gage
115	Length from flange to bottom of primer hole counterbore, max.		0.40%	3.1	Gage
116	Depth of thread relief		0.40%	3.1	Gage
117	Protective coating inadequate or defective		0.40%	3.4	4.5.6

NOTES:
(a) All non-compliance units must be removed from the lot.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63296 (AR)

PARAGRAPH	TITLE	SHEET 2 OF 2	DRAWING NUMBER
4.4.2.2	CONTINUATION SHEET		10520214
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201.	Pitch diameter of thread, min.		0.65%	3.1	Gage
202.	Diameter of primer hole thread relief		0.65%	3.1	Gage
203.	Thread damaged or not full		0.65%	3.8.3	Visual
204.	Body dented or damaged		0.65%	3.1	Visual
205.	Marking missing, incorrect, or illegible		0.65%	3.1	Visual
206.	Wax coating missing		0.65%	3.1	Visual
207.	Evidence of poor workmanship		0.65%	3.8	Visual

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NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Carton - Prior to Sealing	1 OF 1	7548235
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201.	Improper assembly		0.65%	3.1	Visual
202.	Incorrect number in carton		0.65%	3.1	Visual
203.	Separator missing or damaged		0.65%	3.1	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63296 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 7548235
4.4.2.4	Carton - After Sealing				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201.	Carton damaged, cut or gouged through all layers		0.65%	3.1	Visual
202.	Carton improperly sealed		0.65%	3.1	Visual
203.	Contents loose		0.65%	3.1	Visual/Manual
204.	Marking missing, incorrect or illegible		0.65%	3.1	Visual

NOTES:

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4.4.3 Testing.-Testing is described in the First Article and Quality Conformance Inspection tables.

4.4.3.1 Functional tests.-The following proving ground tests will be performed by the Government:

4.4.3.1.1 First article sample.-The quantities listed in Table I will be ballistically tested in the phases indicated and the results of the test will be evaluated in accordance with the criteria set forth in Table II.

TABLE II

DEFECT CHARACTERISTIC	ACCEPT
1. Obturation	0
2. Splits or cracks within one caliber of the head	0
3. Splits or cracks in sidewall more than one caliber from the head	1
4. Extraction	0
5. Primer retention	0

4.4.3.1.2 Production lot.-One hundred cartridge cases from each lot (see 4.4.1), will be ballistically tested in the excess pressure phase. Acceptance and rejection criteria shall be in accordance with Table III. If after the first submission the lot is eligible for retest (did not meet the reject criteria) an additional two hundred samples shall be provided for test. The total cumulative sample of three hundred must now meet the requirements of Table III.

TABLE III

Defect Characteristic	First Submission Accept	First Submission Reject	Cumulative Submission Accept	Cumulative Submission Reject
1. Obturation	0	2	1	2
2. Splits or cracks within one caliber of the head	0	1	Not Permitted	
3. Splits or cracks in sidewall more than one caliber from the head	1	3	2	3
4. Extraction	0	2	1	2
5. Primer retention	0	2	1	2

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4.4.4 Inspection equipment.-The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.8 herein.

4.5 Test methods and procedures

4.5.1 Thermal treatment.-Each cartridge case shall be subjected to a hardness comparator test of a non-destructive type approved by the Contracting Officer.

4.5.2 High stress expansion test.-Each cartridge case in the lot (see 4.4.1) after heat treatment but prior to tapering shall be subjected to this test. The cartridge case shall be expanded diametrically .050 inches (1.27mm) minimum while under pressure in a zone extending from 1.5 inches (38.10mm) above the flange to 9 inches (228.60mm) above the flange, expansion will decrease as the flange is approached, expansion at flange to be zero.

4.5.3 Hardness test.-This test shall be performed on locations specified by Dwg. 10520214 in accordance with the procedures for Rockwell Hardness Testing prescribed by ASTM E-18.

4.5.4 Tension test.-The test specimens shall be taken from the cartridge case in accordance with the standard position and size of the tensile specimen chart described in Figure 1. The yield strength, tensile strength and elongation shall be determined in accordance with the procedures prescribed by ASTM E-8. The results shall be recorded for information only.

4.5.5 Macro examination.-Each sample shall be cut in 90 degree (1.5708 rad) sections along the axis through the primer hole of the sample cartridge case. The specimens shall be etched in accordance with the procedures prescribed for this test by ASTM E381 and visually examined to determine compliance with 3.8.2.

4.5.6 Protective coating.-The criteria specified by the applicable specification shall be utilized in determining the acceptability of the protective coating.

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4.5.6.1 Thickness of coating.-Five cartridge cases from each lot (see 4.4.1) shall be subjected to this test. The thickness of the coating shall be determined by the method specified in the applicable specification. Failure of any specimen to comply with the requirements of the applicable specification shall be cause for rejection of the lot.

4.5.6.2 Salt spray.-Five cartridge cases from each lot (see 4.4.1) shall be subjected to this test. The samples shall be tested using the criteria and equipment specified for this test by ASTM B-117. The inner surface as well as the outer surface of the case will be examined for compliance with the requirements of the applicable specification. Failure of any specimen to comply with the requirements of the applicable drawing and specification shall be cause for rejection of the lot.

4.5.7 Visual Standards.-Each cartridge case shall be visually examined for workmanship and cracks. The cartridge case shall comply with the requirements of 3.8 or established visual standards. The visual standards shall be mutually agreed upon between the government and the contractor.

4.5.7.1 Cracks detection.-Four power (4x) magnification shall be used in the examination of the exterior surface of the sidewall for a minimum distance of eight inches (203.20mm) forward of the base (flange). All other surfaces may be examined without magnification.

4.5.8 Maximum profile and alignment gage.-Each cartridge case, after the application of the protective coating shall freely enter a maximum profile and alignment gage, Dwg. 8663760. The gate of the gage shall close without interference with the cartridge case base (flange).

4.5.9 Functional test

4.5.9.1 First article sample

4.5.9.1.1 Excess pressure phase.-The cartridge case will be assembled into a complete round in the same manner as required for service firing. The propellant charge will be adjusted to produce a chamber pressure with the following limits (see Dwg. 10520214). Minimum mean value of the test rounds fired to be 64,000 psi (441.26 MPa) piezo equivalent pressure and a maximum individual pressure to be 68,500 psi (472.29 MPa) piezo equivalent pressure. The complete rounds will be temperature conditioned at +70°F ± 2.5°F (+21°C ± 1.5°C) and fired in a serviceable weapon for which the round is standard. The weapon will be operated in the service manner.

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4.5.9.1.2 Cold temperature phase.-The cartridge case will be assembled into a complete round in the same manner as required for service firing. The propellant charge will be adjusted to produce a chamber pressure of 57,100 psi (393.69 MPa) to 60,600 psi (417.82 MPa) piezo equivalent pressure, when temperature conditioned at 70°F. The complete rounds will be temperature conditioned at -35°F + 5°F (-37°C + 3°C) and fired in a serviceable weapon for which the round is standard. The weapon will be operated in the service manner.

4.5.9.1.3 Excess, excess pressure phase.-The cartridge case will be assembled into a complete round in the same manner as required for service firing. The propellant charge will be adjusted to produce a chamber pressure of 72,300 psi (498.49 MPa) to 74,200 psi (511.59 MPa) piezo equivalent pressure. The complete rounds will be temperature conditioned at +70°F ± 2.5°F (+21°C ± 1.5°C) and fired in a serviceable weapon for which the round is standard. The weapon will be operated in the service manner.

4.5.9.2 Production lot

4.5.9.2.1 Excess pressure phase.-See 4.5.9.1.1 herein.

5. PACKAGING

5.1 Packaging requirements.-Packaging shall be as specified on Dwg. 7548235.

5.2 Packing and marking.-Packing and marking shall be as specified on Dwg. 7548235 and MIL-STD-1169.

6. NOTES

6.1 Intended use.-This cartridge case is intended to be used in the 105MM, APDS-T and TPDS-T type ammunition.

6.2 Manufacturing test lots.-Prior to fabrication of a first article sample, the contractor upon approval of the contracting officer, may make and submit for evaluation up to five manufacturing test lots. These manufacturing test lots shall consist of not less than ten nor more than twenty-five cartridge cases each. The contractor shall furnish all information pertaining to the manufacture of the lots to the contracting officer. All costs for testing these process control lots will be borne by the Government.

6.3 Visual examination qualification.-When compliance with the applicable requirement is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

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6.4 Extraction.-The term "extraction" shall mean that the fired cartridge is loosened and pulled to the rear of the weapon sufficiently to permit removal by hand.

6.5 Ordering data.-See MIL-A-48078.

6.6 Delivery of data cards (prepared in accordance with MIL-STD-1167) to the technical activity.-The following address is to be added to those listed in the contract (Form DD 1423) for delivery of one copy, each, to the technical activity, of the Ammunition Data Cards and Quality Inspection Defects Reports prepared by the contractor for the duration of the contract: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-E, Dover, New Jersey 07801.

6.7 Firing record information.-The following addresses are to be added to the Ballistic Test Request (BTR) for delivery of firng test reports to the technical activity: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-E and DRDAR-LCU-M, Dover, New Jersey 07801.

6.8 Submission of inspection equipment designs for approval.-See MIL-A-48078. Submit equipment designs, as required to:

Commander
US Army Research and Development Command
ATTN: DRDAR-QAR-I
Dover, NJ 07801

6.9 Supersession data.-This specification includes the requirements of Purchase Description FA-PD-MI-2523, Rev. 1, dated 1 July 1964.

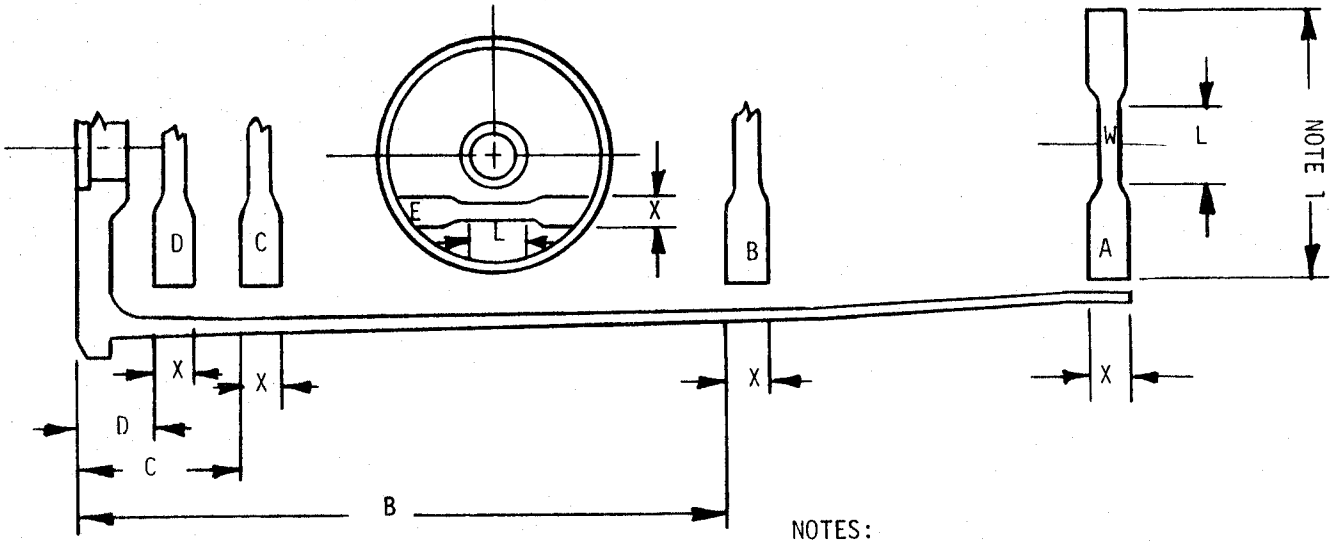
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STANDARD POSITIONS AND SIZES OF TENSILE
SPECIMENS FOR CASE, CARTRIDGE, 105MM, M115B1



SPECIMEN	X	W	L	LOCATION FROM HEAT
A	.40 IN (10.16MM)	.25 IN (6.35MM)	1.0 IN (25.4MM)	AT MOUTH
B	.40 IN (10.16MM)	.25 IN (6.35MM)	1.0 IN (25.4MM)	15.25 IN (387.25MM)
C	.40 IN (10.16MM)	.25 IN (6.35MM)	1.0 IN (25.4MM)	2.00 IN (50.80MM)
D	.40 IN (10.16MM)	.25 IN (6.35MM)	1.0 IN (25.4MM)	0.75 IN (19.05MM)
E	.40 IN (10.16MM)	.25 IN (6.35MM)	1.0 IN 25.4MM)	IN HEAD

NOTES:

1. OVERALL LENGTH OF SPECIMEN TO SUIT FIXTURE IN TESTING MACHINE.
2. SPECIMEN C AND D TO BE MACHINED TO UNIFORM THICKNESS PARALLEL TO INSIDE SURFACE.
3. CENTERLINE OF SPECIMENT E TO BE 1.75 IN (44.45MM) FROM CENTERLINE OF CARTRIDGE CASE.
4. SPECIMEN E TO BE MACHINED SO AS TO ELIMINATE EFFECTS OF STAMPING ON TEST.
5. YIELD STRENGTH IN 0.2 PERCENT OFFSET.
6. PERCENT ELONGATION IN 1.0 IN (25.4MM) GAGE LENGTH.

FIGURE I

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-63296 CASE, CARTRIDGE, 105MM, M115B1

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

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